CERTAIN-COAST — brief for a partner-country team

Certified early warning to protect water, port and energy infrastructure on a sinking Adriatic coast

NATO Science for Peace and Security (SPS) Programme · Multi-Year Project · call 2027-1 · submission by 15 November 2026, 23:59 CET · coordinated by the Faculty of Geology and Mining, Polytechnic University of Tirana, Albania

The project in brief

Near Durrës, the main port of Albania, the wellfields that supply drinking water, the port and an energy landfall stand on soft ground that is slowly sinking, close to active faults (the Mw 6.4 earthquake of 26 November 2019 struck there). Three hazards threaten them together: seawater entering the wells, liquefaction in an earthquake, and coastal flooding in storms. Sinking land and rising seas worsen all three. CERTAIN-COAST builds one model of the three hazards, a fast machine-learned copy of it whose error is bounded for every forecast, and warnings that state in advance how often they may miss a dangerous event. The warnings run live for twelve months with the water utility, the port, the energy operator and civil protection, and are then handed over to them. **36 months; about €382,500 requested from NATO.**

Why it fits the call

The 2027-1 call names among its topics of particular interest "protection of critical infrastructure, supplies and personnel; maritime and undersea infrastructure resilience". CERTAIN-COAST protects the water, port and energy infrastructure of a NATO Ally, and the method transfers to other subsiding coasts of Allies and partners, which is where the partner country comes in.

What is new

- **Warnings with a stated miss rate**, verified during live operation with a test that stays valid however often the record is checked.
- **A certified safe operating window**: the pumping and dewatering decisions that keep all three hazards within limits at once, and the sea level or year at which this window closes. In our literature search we found no published method for this.
- **The value of a sensor in years of safe operation**, used to choose the monitoring network.

The role we propose for the partner-country team

Work package	What the partner team leads
WP2 Coupled model and certified surrogate	Coupled model of intrusion, compaction, liquefaction and sea level; calibration; a neural-operator or kernel surrogate with an error bound on every forecast
WP3 Certified multi-hazard alarm	Alarm margins and their proofs; a joint guarantee over the three alarms; back-testing on the record and on storylines beyond it
WP6 Training, transfer and dissemination	Training of early-career researchers; test of the method on a subsiding coast of the partner country; open software and data

The NATO-country team (Albania) leads coastal motion, the monitoring network, live operation and management (WP1, WP4, WP5, WP7). Joint papers, co-supervision of early-career researchers and shared open software are planned throughout.

Budget for the partner institution in the current plan: €84,000 (22% of the total)

Item	Partner share	Basis
Computing server with GPU	€12,000	partner
Software licences	€3,000	partner
Consortium meetings (kick-off, mid-term, final): travel and subsistence	€10,800	shared equally
Young-researcher exchanges, two months each	€21,000	partner
Conference presentations	€7,200	shared equally
Stipends for young researchers without a PhD: 4 researchers × 30 months × €400	€24,000	shared equally
Open-access publication fees	€6,000	shared equally

The share can be raised, for example by placing more stipends or the transfer study at the partner institution; we will agree the final figures together before submission.

Who can be the partner-country Project Director

- Employed and resident in an eligible partner country of the SPS Programme, at a university, public research institute or other non-profit institution (companies cannot be co-directors).
- Holds no other SPS grant at the time (one SPS grant per person).
- Gives about 30% of working time to the project; co-signs the SPS Agreement for Joint Research with a representative of the institution.

What we need, and by when

By	What
Thursday 5 November	A reply: interest in principle, the name of the proposed Project Director and institution
Monday 9 November	Details for the application form: nationality, position and institution address, employment history and education, recent publications, research focus (up to 500 words), other grants; a one-page overview of the institution (capabilities, resources, facilities, websites)
Thursday 12 November	The signed SPS MYP Agreement for Joint Research (NATO template, which we send)
Sunday 15 November	Submission by the NATO-country Project Director

The full proposal text is shared as soon as you confirm interest. Contact: Dulian Zeqiraj, Faculty of Geology and Mining, Polytechnic University of Tirana (please reply to the e-mail that brought this brief).